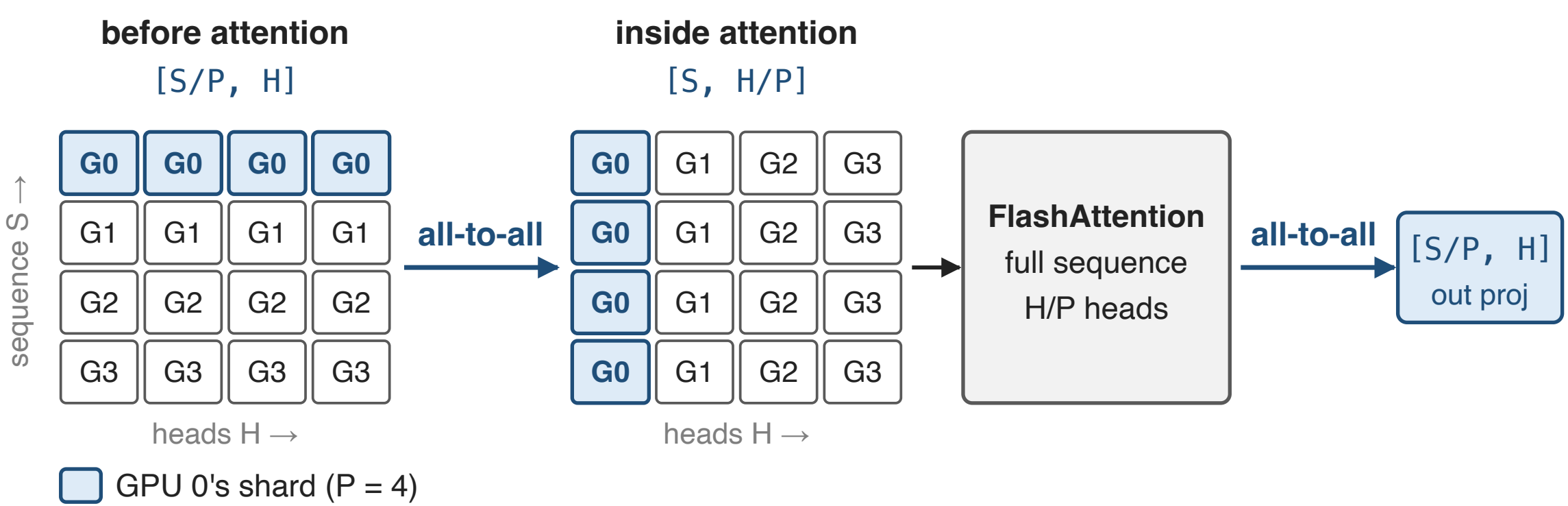
before attention
[S/P, H]
inside attention
[S, H/P]
sequence S →
G0 G0 G0 G0
G1 G1 G1 G1
G2 G2 G2 G2
G3 G3 G3 G3
heads H →
all-to-all
G0 G1 G2 G3
G0 G1 G2 G3
G0 G1 G2 G3
G0 G1 G2 G3
heads H →
FlashAttention
full sequence
H/P heads
all-to-all
[S/P, H]
out proj
GPU 0's shard (P = 4)